

ANDREW LUO

PhD Candidate (Computer Science), University of Waterloo

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EDUCATION

University of Waterloo

Doctor of Philosophy - Computer Science

Waterloo, Ontario

Jan 2024 - Present

- Conducting research in symbolic computational methods for Analytic Combinatorics in Several Variables under the supervision of Stephen Melczer and Eric Schost.
- **Courses:** Computational Complexity, Algebraic Geometry, Algebraic Complexity, Artificial Intelligence, Software Design and Architecture, Computer Networks, Complexity in Commutative Algebra, University Mathematics Teaching Techniques, Algorithmic Spectral Graph Theory

University of Waterloo

Bachelor of Mathematics

Waterloo, Ontario

Sep 2018 - Apr 2023

- Triple majors in Computer Science, Pure Mathematics, and Combinatorics & Optimization.
- **Courses:**
 - PMATH – Groups and Rings, Fields and Galois Theory, Real Analysis, Complex Analysis, Differential Geometry, Analytic Number Theory, Lebesgue Integration and Fourier Analysis, Algebraic Number Theory, Representation of Finite Groups, Set Theory and Model Theory
 - CS – Functional Programming, Object Oriented Programming, Data Structures and Algorithms, Numerical Computation, Theory of Computation, Algorithm Design and Analysis, Machine Learning
 - CO – Intro to Optimization (Advanced Level), Combinatorial Enumeration, Applied Cryptography, Mathematics of Public Key Cryptography, Algebraic Combinatorics, Continuous Optimization

PROFESSIONAL DEVELOPMENT

Certificate in University Teaching

University of Waterloo

Waterloo, ON

May 2026 - Present

- Enrolling in comprehensive program offered by the Center of Teaching Excellence for developing skills in teaching and learning in higher level education.

Fundamentals of University Teaching

University of Waterloo

Waterloo, ON

Feb 2026 - Present

- Enrolling in program offered Center of Teaching Excellence for developing skills in lesson planning, teaching, teaching assistance.
- Completed workshops in Effective Lesson Plans, Teaching Methods, Building Rapport With Students, Building TA-Instructor Rapport, and Teaching Stem Tutorials.
- Currently completed two of the three required micro-teaching sessions.

Math 900

University of Waterloo

Waterloo, ON

Jan - Apr 2026

- Attended lectures in Math 900: University Mathematics Teaching Techniques which covered topics ranging from presentation skills, active learning methods, types of assessments, and handling classroom scenarios.
- Completed three 15-minute micro-teaching sessions and a teaching practicum.

PRESENTATIONS

Applications of Computer Algebra in Analytic Combinatorics, Symbolic Computation Group Seminar 2026, University of Waterloo	Waterloo, ON
A SageMath Package for Analytic Combinatorics in Several Variables, SageDays 2025, Hokkaido University	Sapporo, Japan
A SageMath Package for Analytic Combinatorics in Several Variables: Beyond the Smooth Case (Code Demo), FPSAC 2025, Hokkaido University	Sapporo, Japan
Automatically Computing Asymptotics of Sequences with Multivariate Rational Generating Functions, AofA 2025, Fields Institute	Toronto, ON
A Brief Introduction to Computational Analytic Combinatorics, Canadian Undergraduate Mathematics Conference, Université Laval	Québec City, QC
Derangements, The Exponential Function, and The Principle of Inclusion Exclusion, Fleetwood Park Secondary School Math Club	Surrey, BC
An Introduction to Vinogradov's Mean Value Theorem, Canadian Undergraduate Mathematics Conference, Western University	London, ON

PUBLICATIONS

A SageMath Package for Analytic Combinatorics in Several Variables: Beyond the Smooth Case. B. Hackl, A. Luo, S. Melczer, E. Schost. Submitted March 2025.	FPSAC Link
Rigorous Analytic Combinatorics in Several Variables in SageMath. B. Hackl, A. Luo, S. Melczer, J. Selover, and E. Wong. Submitted March 2023.	FPSAC Link

SOFTWARE

sage-acsv: a Sage package for the study of the asymptotic behaviour of multivariate rational generating functions using symbolic-numeric representations of algebraic varieties.	GitHub Link
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TEACHING EXPERIENCE

Instructional Apprentice - CS 240 <i>University of Waterloo</i> Holding weekly interactive tutorials and assisting on class forums.	Waterloo, ON <i>May 2026 - Aug 2026</i>
Teaching Assistant - AMATH 242 / CS 371 <i>University of Waterloo</i> Assisted with marking of assignments and exams, creating solution videos, responding on the class forum, conducting office hours, and proctoring exams.	Waterloo, ON <i>Jan - Apr 2026</i>
Guest Lecturer - CS 116 <i>University of Waterloo</i> Delivered an in-person guest lecture for CS 116: Introduction to Computer Science 2 on the topic of sorting and dictionaries as part of the teaching practicum requirement for Math 900.	Waterloo, ON <i>Mar 2026</i>
Teaching Assistant - CS 370 <i>University of Waterloo</i> Assisted with marking of assignments and exams, giving tutorials on Python and Jupyter, responding on the class forum, conducting office hours, and proctoring exams.	Waterloo, ON <i>Jan - Aug 2024, Jan - Apr 2025</i>

Instructional Apprentice - CS 245*University of Waterloo*

Waterloo, ON

Sep - Dec 2024

- Conducted weekly tutorials and office hours. Assisted with proctoring of exams.

Undergraduate Teaching Assistant - Math 235*University of Waterloo*

Waterloo, ON

Jan 2021 - Aug 2022

- Assisted with grading of assignments.

Undergraduate Teaching Assistant - Math 136*University of Waterloo*

Waterloo, ON

Sep 2019 - Aug 2020

- Assisted with grading of weekly quizzes.

SERVICES AND OUTREACH**Canada/USA Mathcamp QQ Math Jam Presenter***American Mathematical Society*

Remote

May 2026

- Presented a solution for the Mathcamp Qualifying Quiz in the Art of Problem Solving online classroom.

SPARCS Workshop Assistant*Center of Education for Mathematics and Computing*

Waterloo, ON

May 2026

- Assisted students with week-long interactive programming workshop.

Directed Reading Project Mentor*University of Waterloo*

Waterloo, ON

Jan - Apr 2026

- Mentored three undergraduate students on a part-time mathematics reading group project organized by the Women in Mathematics committee.
- Guided mentees on the study of algorithms for polynomial system solving.

Canada/USA Mathcamp Admissions Grader*American Mathematical Society*

Cambridge, MA (Remote)

March 2026

- Assisted in the grading of the Qualifying Quiz portion of admissions for Canada/USA Mathcamp.

Euclid Math Contest Marker*Center of Education for Mathematics and Computing*

Waterloo, ON

Apr 2024, 2025

- Assisted in the grading of Euclid Math Contest paper submissions.

SPARCS Volunteer*Center of Education for Mathematics and Computing*

Waterloo, ON

May 2024

- Assisted workshop speakers with set-up and answering student questions.

RESEARCH EXPERIENCE**Undergraduate Mathematics Research Assistant***University of Waterloo*

Waterloo, ON

May - Aug 2022

- Studied the mathematical background of Analytic Combinatorics in Several Variables under the supervision of Stephen Melczer.
- Worked alongside S. Melczer, B. Hackl, J. Selover, E. Wong to develop the first software package in SageMath for rigorously computing asymptotics of multivariate generating functions with smooth singular varieties.

Undergraduate Mathematics Research Assistant*University of Waterloo*

Waterloo, ON

May - Aug 2021

- Surveyed the most recent progress made in and applications to the asymptotic results of Waring's Problem and Vinogradov's Mean Value Theorem under supervision of Professor Yu-Ru Liu.

PROFESSIONAL EXPERIENCE

Mathematics Tutor

Cloudtimes Education

Mississauga, ON

Feb 2023 - June 2024

- Helped students with course and homework review in high school math topics.
- Developed and delivered notes, content, and review questions for Euclid Math Contest preparation.

Full-Stack Software Developer Intern

Noom

New York City, NY

Jan - Apr 2022

- Used the Apache OpenNLP library to develop a Natural Language Processing pipeline for processing and triage of user messages and help requests.
- Improved question detection logic by 15%, reducing the wait times of customer service requests and improving long-term user retention by 5%.

Full-Stack Developer Intern

BDO

Mississauga, ON

Sep - Dec 2020

- Developed an interactive scheduling web app in Angular, .NET, and Microsoft SQL Server for visualizing and modifying allocation of resources.

Backend Developer Intern

Wayfair

Boston, MA

Jan - Apr 2020

- Implemented backend business logic using PHP and Microsoft SQL Server, ensuring correctness by creating extensive unit and integration tests covering all cases of the business logic.

Full-Stack Software Developer Intern

Wish

San Francisco, CA

May - Aug 2019

- Wrote Python and SQL scripts to monitor merchant fraud lifecycle.
- Extensively refactored legacy code and eliminated large amounts of technical debt.

HONOURS AND AWARDS

Canada Graduate Scholarship - Master's NSERC \$27000	2025
President's Graduate Scholarship University of Waterloo \$10000	2025
Queen Elizabeth II Scholarship in Science & Technology University of Waterloo \$15000	2024
President's Graduate Scholarship University of Waterloo \$5000	2024
Graduate Excellence Award in Computer Science University of Waterloo \$5000	2024
Mike Vangoch Memorial Scholarship University of Waterloo \$1200	2023
NSERC Undergraduate Student Research Award NSERC \$6000	2022
President's Research Award University of Waterloo \$1500	2021
NSERC Undergraduate Student Research Award NSERC \$6000	2021
President's International Experience Award University of Waterloo \$1500	2021
Faculty of Mathematics Entrance Scholarship University of Waterloo \$5000	2018
Putnam Math Competition Honour Roll Mathematical Association of America Top 500	2018